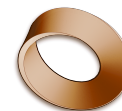


Name: _____

Date: _____

Mr. Rawson • Y8 MYP Design



1.1 – Introduction to MakeCode

Unit 1: Robotics & Control

1. **Open the editor.** Go to `makecode.microbit.org` → **New Project** → name it name-badge. Name it *now*, not later.
2. **First frame.** From `Basic`, drag `show leds` inside `on start`. Click the squares to draw the first letter of your name. The simulator on the left shows it straight away.
3. **Onto the board.** Plug the micro:bit in with the USB cable, then click the purple **Download** button. If a file lands in your Downloads folder, drag it onto the **MICROBIT** drive — exactly like copying to a memory stick. *Your board should now show the letter. Do not go on until it does.*
4. **Spell your name.** Add one `show leds` frame per letter, and after each frame add `pause (ms) 500`. Without the pauses the letters blur past.
5. **Make it cycle.** From `Loops`, wrap all your blocks inside `repeat 4 times`. Download again and watch it loop.
6. **Add an input.** From `Input`, drag in `on button A pressed`. Move *all* your blocks into it, out of `on start`. Now nothing happens until you press A: button = input, your loop = process, the LEDs = output.
7. **Publish it. Share** → **Publish project** → copy the link and write it below. The link is case-sensitive and cannot be searched for later — this link is how you will submit your homework.

My sharing link: _____

Stuck? Check these first

- **Simulator works, board blank** — is the cable clicked in at both ends? Did the download finish? Do step 3 again, slowly.
- **Letters flash too fast (or never change)** — change the pause number. 1000 ms = 1 second.
- **A letter will not fit** — the grid is only 5×5 . Straight-line letters (E, L, T, I) are easiest; initials are fine.
- **Where did my old program go?** — the micro:bit holds *one* program at a time. A new download replaces the old one. That is normal.
- **Lost the link** — Share → Publish again. You get a new link to the same program.

Extension

Finished? Make `on button B pressed` do something different • shrink the pauses until your name is unreadable, then find the readable limit • try `show string` with your name — what must it be doing underneath that you did by hand?